PONMATHISH T

DAY - 07

1. What is a network address?

A network address is an IP address that identifies a particular network. It is used to represent the network itself, not a specific device. For example, in the network 192.168.1.0/24, **192.168.1.0** is the network address.

2. What is a broadcast address?

A broadcast address is used to send data to all devices within a particular network. For example, in a 192.168.1.0/24 network, **192.168.1.255** is the broadcast address.

3. What is a usable host address?

A usable host address is an IP address that can normally be assigned to a device such as a computer, phone, or printer on a network. It is neither the network address nor the broadcast address.

4. What is the first usable IP address?

The first usable IP address is normally the address immediately after the network address. For example, in **192.168.1.0/24**, the first usable address is **192.168.1.1**.

5. What is the last usable IP address?

The last usable IP address is normally the address immediately before the broadcast address. For **192.168.1.0/24**, the last usable address is **192.168.1.254**.

6. What does /24 mean?

The **/24** notation means that the first 24 bits of the IPv4 address are used for the network portion. The remaining 8 bits are available for hosts.

7. What subnet mask corresponds to /24?

The subnet mask for **/24** is **255.255.255.0**. It provides 24 network bits and 8 host bits.

8. What is a default gateway?

A default gateway is normally a router that allows devices on a local network to communicate with other networks. For example, a home router may act as the default gateway for computers and phones.

9. Why does a router interface need an IP address?

A router interface needs an IP address so that it can communicate with devices on the connected network and route traffic between different networks. The IP address identifies that interface on the network.

10. What is a static IP configuration?

A static IP configuration is when the IP address and other network settings are manually assigned to a device. The address normally stays the same unless an administrator changes it. Static IPs are commonly used for servers and network devices.

11. What does "show ip interface brief" display?

The Cisco command **show ip interface brief** provides a short summary of the router's interfaces. It commonly shows the interface name, IP address, and the interface and line status, such as **up/up**.

12. What does "no shutdown" do conceptually on a Cisco interface?

The **no shutdown** command administratively enables a Cisco interface. If the interface is physically and logically working correctly, it can then become operational and show an **up/up** status.

13. What does interface status "up/up" generally indicate?

An **up/up** status generally means that the interface is enabled and its physical and data-link connections are working. In simple terms, the interface is operational.

14. Why should network configuration always be verified?

Network configuration should be verified to make sure that the IP address, subnet mask, gateway, and other settings are correct. Verification helps identify configuration mistakes and connectivity problems before they affect users or services.

15. Why is IPv4 configuration important in cybersecurity?

IPv4 configuration is important in cybersecurity because IP addresses help identify devices and control network communication. Understanding IP addresses, subnets,

gateways, and interfaces helps security professionals monitor traffic, configure firewalls, detect suspicious connections, and investigate security incidents.

IPv4 EXERCISE

Q1. For 192.168.100.0/24, what is the Network Address?

Answer: The Network Address is **192.168.100.0**.

Q2. For 192.168.100.0/24, what is the First Usable IP?

Answer: The First Usable IP address is **192.168.100.1**.

Q3. For 192.168.100.0/24, what is the Last Usable IP?

Answer: The Last Usable IP address is **192.168.100.254**.

Q4. For 192.168.100.0/24, what is the Broadcast Address?

Answer: The Broadcast Address is **192.168.100.255**.

PRACTICAL

Q5. What is the IPv4 Address of your PC?

Answer: 10.150.34.91

Q6. What is the Subnet Mask of your PC?

Answer: 255.255.255.0

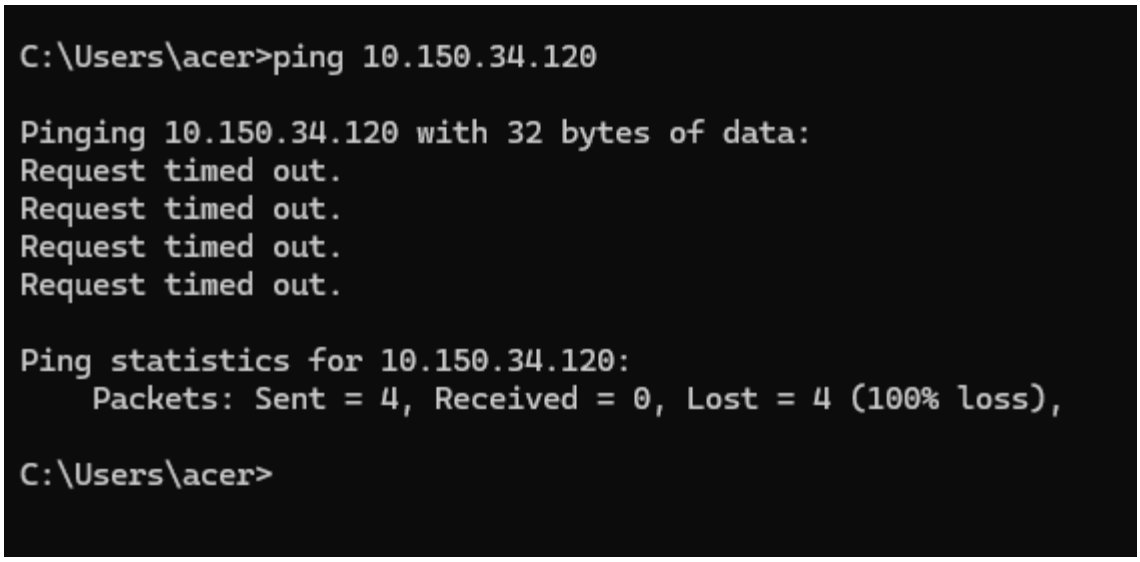
Q7. What is the Default Gateway of your PC?

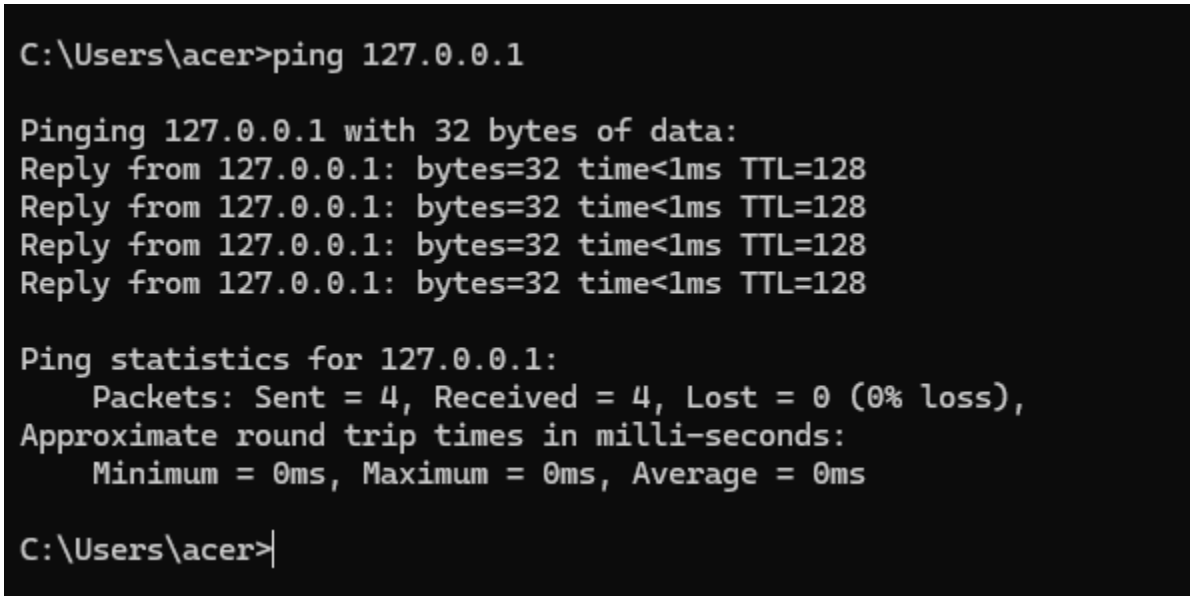
Answer: 10.150.34.120

Q8. What is the DNS Server of your PC?

Answer: 10.150.34.120

SCREENSHOTS





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C:\Users\acer>ipconfig/all

Windows IP Configuration

   Host Name . . . . . . . . . . . . : PM
   Primary Dns Suffix  . . . . . . . :
   Node Type . . . . . . . . . . . . : Hybrid
   IP Routing Enabled. . . . . . . . : No
   WINS Proxy Enabled. . . . . . . . : No

Wireless LAN adapter Local Area Connection* 8:

   Media State . . . . . . . . . . . : Media disconnected
   Connection-specific DNS Suffix  . :
   Description . . . . . . . . . . . : Microsoft Wi-Fi Direct Virtual Adapter
   Physical Address. . . . . . . . . : 5A-E[redacted]
   DHCP Enabled. . . . . . . . . . . : Yes
   Autoconfiguration Enabled . . . . : Yes

Wireless LAN adapter Local Area Connection* 9:

   Media State . . . . . . . . . . . : Media disconnected
   Connection-specific DNS Suffix  . :
   Description . . . . . . . . . . . : Microsof[redacted]irect Virtual Adapter #2
   Physical Address. . . . . . . . . : DA-E[redacted]
   DHCP Enabled. . . . . . . . . . . : Yes
   Autoconfiguration Enabled . . . . : Yes

Wireless LAN adapter Wi-Fi:

   Connection-specific DNS Suffix  . :
   Description . . . . . . . . . . . : Realtek 8821CE Wireless LAN 802.11ac PCI-E NIC
   Physical Address. . . . . . . . . : 58-E4-EB-41-04-F2
   DHCP Enabled. . . . . . . . . . . : Yes
   Autoconfiguration Enabled . . . . : Yes
   IPv6 Address. . . . . . . . . . . : 2409:40f4:[redacted]c(Preferred)
   Temporary IPv6 Address. . . . . . : 2409:40f4:10[redacted]55(Preferred)
   Link-local IPv6 Address . . . . . : fe80::88e[redacted]referred)
   IPv4 Address. . . . . . . . . . . : 10.150.94.94(Preferred)
   Subnet Mask . . . . . . . . . . . : 255.255.255.0
   Lease Obtained. . . . . . . . . . : 29 October 2026 07:52:08
   Lease Expires . . . . . . . . . . : 29 October [redacted]
   Default Gateway . . . . . . . . . : fe80::bc8d:[redacted]
                                       10.150.[redacted]
   DHCP Server . . . . . . . . . . . : 10.150.[redacted]
   DHCPv6 IAID . . . . . . . . . . . : 173597931
   DHCPv6 Client DUID. . . . . . . . : 00-01-00-01-30-A0-86-E6[redacted]
   DNS Servers . . . . . . . . . . . : 10.15[redacted]
                                       2409[redacted]
   NetBIOS over Tcpip. . . . . . . . : Enabled

Ethernet adapter Bluetooth Network Connection:

   Media State . . . . . . . . . . . : Media disconnected
   Connection-specific DNS Suffix  . :
   Description . . . . . . . . . . . : Bluetooth Device (Personal Area Network)
   Physical Address. . . . . . . . . : 58-E[redacted]
   DHCP Enabled. . . . . . . . . . . : Yes
   Autoconfiguration Enabled . . . . : Yes
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